MTH 200 Chapter 4: Applications of Derivatives

4.8 L'Hopital's Rule; Indeterminate Forms

Motivation: How could we evaluate the following limits?

$$\lim_{x\to 1}\frac{\ln(x)}{x-1} \qquad \lim_{x\to\infty}\frac{\ln(x)}{x-1}$$

Notice that if we tried to use direct substitution for these limits we would get:

$\lim_{x\to 1}\frac{\ln(x)}{x-1}\to\frac{\ln(1)}{1-1}=\frac{0}{0}$, which is what we call an <u>indeterminate form</u>.

$\lim_{x\to\infty}\frac{\ln(x)}{x-1}\to\frac{\infty}{\infty}$, which is another type of <u>indeterminate form</u>.

Simplification of these limits seems difficult because:

- We cannot factor. There are no fractions to combine or split. Multiplying by the conjugate is unhelpful.
- Checking the limit from the left and right is not helpful because we still obtain $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
- The Squeeze Theorem is difficult to use.

The idea we will discuss in this section will help us evaluate limits that have an indeterminate form.

Definition (Intuitive): When direct substitution in a limit results in any of the following forms:

$$\frac{0}{0} \qquad \frac{\pm\infty}{\pm\infty} \qquad 0\times\pm\infty \qquad \infty-\infty \qquad 1^{\infty} \qquad 0^{0} \qquad \infty^{0}$$

we say the limit is in an **indeterminate form**.

Theorem (L'Hopital's Rule): Suppose *f* and *g* are differentiable and $g'(x)\neq 0$ on an open interval that contains *a* (except possibly at *a*) and. Suppose that $\lim_{x\to a}\frac{f(x)}{g(x)}$ is in the indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$., then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

provided that the limit on the right exists or is infinite.

Note: L'Hopital's rule still applies if a is replaced with infinity or negative infinity. It also applies to right and left hand limits.

Proof: *(special case only)*

Suppose that $\lim_{x\to a}\frac{f(x)}{g(x)}$ is in the indeterminate form $\frac{0}{0}$, $f'(x)$ & $g'(x)$ are continuous at $x=a$ and $g'(x)\neq 0$.

Since $\lim_{x\to a}\frac{f(x)}{g(x)}$ is in the indeterminate form $\frac{0}{0}$ then $f(a)=0$ & $g(a)=0$.

Note that since $f'(x)$ & $g'(x)$ are continuous at $x=a$ this means that *f* and *g* are differentiable at *a* and therefore are continuous at *a*. Therefore we know that: $\lim_{x\to a} f(x)=f(a)=0$ & $\lim_{x\to a} g(x)=g(a)=0$.

Now we can say:

$$\lim_{x\to a}\frac{f'(x)}{g'(x)}=\frac{f'(a)}{g'(a)}=\frac{\lim_{x\to a}\frac{f(x)-f(a)}{x-a}}{\lim_{x\to a}\frac{g(x)-g(a)}{x-a}}=\lim_{x\to a}\frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}}=\lim_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)}=\lim_{x\to a}\frac{f(x)}{g(x)}$$

Example 1: Find each of the following limits.

a) $\lim_{x\to 1}\frac{\ln(x)}{x-1}$

We notice that this limit has the indeterminate form of $\frac{0}{0}$. Therefore by L'Hopital's rule we have:

$$\lim_{x\to 1}\frac{\ln(x)}{x-1}\overset{H}{=}\lim_{x\to 1}\frac{\left(\frac{1}{x}\right)}{1}=\lim_{x\to 1}\frac{1}{x}=1 .$$

Note: We place "H" above the equal sign to indicate that we used L'Hopital's rule. Be sure that you have checked to see whether the conditions of L'Hopital's rule are satisfied before using this rule.

b) $\lim_{x\to \infty}\frac{\ln(x)}{x-1}$

We notice that this limit has the indeterminate form of $\frac{\infty}{\infty}$. Therefore by L'Hopital's rule we have:

$$\lim_{x\to \infty}\frac{\ln(x)}{x-1}\overset{H}{=}\lim_{x\to \infty}\frac{\left(\frac{1}{x}\right)}{1}=\lim_{x\to \infty}\frac{1}{x}=0 .$$

Example 2: Determine each of the following limits.

a) $\lim\limits_{x\to 0}\dfrac{\sin(x)}{x}$

We notice that this limit has the indeterminate form of $\dfrac{0}{0}$. Therefore by L'Hopital's rule we have:

$$\lim_{x\to 0}\frac{\sin(x)}{x}\overset{H}{=}\lim_{x\to 0}\frac{\cos(x)}{1}=\lim_{x\to 0}\cos(x)=1.$$

Note: We may recall that we determined this limit using a complicated geometric argument back in section 3.5. Be careful to note though that without the geometric argument from 3.5 we would not be able to use L'Hopital's rule since we wouldn't know how to compute the derivative of $\sin(x)$*!*

b) $\lim\limits_{x\to\infty}\dfrac{\ln(x)}{2\sqrt{x}}$

We notice that this limit has the indeterminate form of $\dfrac{\infty}{\infty}$. Therefore by L'Hopital's rule we have:

$$\lim_{x\to\infty}\frac{\ln(x)}{2\sqrt{x}}\overset{H}{=}\lim_{x\to\infty}\frac{\left(\frac{1}{x}\right)}{\frac{2}{2\sqrt{x}}}=\lim_{x\to\infty}\frac{\left(\frac{1}{x}\right)}{\left(\frac{1}{\sqrt{x}}\right)}=\lim_{x\to\infty}\frac{\sqrt{x}}{x}=\lim_{x\to\infty}\frac{1}{\sqrt{x}}=0.$$

c) $\lim\limits_{x\to\infty}\dfrac{e^{3x}}{x^2+3x+5}$

We notice that this limit has the indeterminate form of $\dfrac{\infty}{\infty}$. Therefore by L'Hopital's rule we have:

$$\lim_{x\to\infty}\frac{e^{3x}}{x^2+3x+5}\overset{H}{=}\lim_{x\to\infty}\frac{3e^{3x}}{2x+3}\overset{H}{=}\lim_{x\to\infty}\frac{9e^{3x}}{2}=\infty.$$

Example 3: Determine what is wrong with each of the following limit calculations.

a) $\lim\limits_{x\to\pi}\dfrac{\cos(x)}{x-\pi}\overset{H}{=}\lim\limits_{x\to\pi}\dfrac{-\sin(x)}{1}=-\sin(\pi)=0$

The application of L'Hopital's rule is not possible because the limit is not in an indeterminate form!

b) $\lim\limits_{x\to 0}\dfrac{3x^2}{x^2+2x}\overset{H}{=}\lim\limits_{x\to 0}\dfrac{6x}{2x+2}\overset{H}{=}\lim\limits_{x\to 0}\dfrac{6}{2}=3$

Here the second application of L'Hopital's rule does not work because the limit was again not in an indeterminate form!

Note: This example is very important because it illustrates that checking whether the limit is of a form to which L'Hopital's rule does indeed apply is paramount.

Example 4: Find $\lim_{x\to 0^+} x\ln(x)$.

We notice that this limit has the indeterminate form of $0\times\infty$. However we cannot yet apply L'Hopitals rule because we first need the indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We note that $\lim_{x\to 0^+} x\ln(x) = \lim_{x\to 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)}$. This limit has the indeterminate form of $\frac{-\infty}{\infty}$, so we can apply L'Hopital's rule.

$$\lim_{x\to 0^+} x\ln(x) = \lim_{x\to 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)} \overset{H}{=} \lim_{x\to 0^+} \frac{\left(\frac{1}{x}\right)}{\left(\frac{-1}{x^2}\right)} = \lim_{x\to 0^+} \frac{1}{x}\cdot\left(-\frac{x^2}{1}\right) = \lim_{x\to 0^+} -x = 0$$

Example 5: Find $\lim_{x\to -\infty} xe^x$.

We notice that this limit has the indeterminate form of $-\infty\times 0$. However we cannot yet apply L'Hopitals rule because we first need the indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We note that $\lim_{x\to -\infty} xe^x = \lim_{x\to -\infty} \frac{x}{e^{-x}}$. This limit has the indeterminate form of $\frac{-\infty}{\infty}$, so we can apply L'Hopital's rule.

$$\lim_{x\to -\infty} xe^x = \lim_{x\to -\infty} \frac{x}{e^{-x}} \overset{H}{=} \lim_{x\to -\infty} \frac{1}{-e^{-x}} = \lim_{x\to \infty} \frac{1}{-e^{x}} = 0$$

Example 6: Find $\lim_{x\to 0^+} \left(\frac{1}{\sin(x)} - \frac{1}{x}\right)$.

We notice that this limit has the indeterminate form of $\infty-\infty$. However we cannot yet apply L'Hopitals rule because we first need the indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We note that $\lim_{x\to 0^+} \left(\frac{1}{\sin(x)} - \frac{1}{x}\right) = \lim_{x\to 0^+} \frac{x-\sin(x)}{x\sin(x)}$. This limit has the indeterminate form of $\frac{0}{0}$, so we can apply L'Hopital's rule.

$$\lim_{x\to 0^+} \left(\frac{1}{\sin(x)} - \frac{1}{x}\right) = \lim_{x\to 0^+} \frac{x-\sin(x)}{x\sin(x)} \overset{H}{=} \lim_{x\to 0^+} \frac{1-\cos(x)}{\sin(x)+x\cos(x)} \overset{H}{=} \lim_{x\to 0^+} \frac{\sin(x)}{\cos(x)+\cos(x)-x\sin(x)} = \frac{0}{1+1-0} = \frac{0}{2} = 0$$

Dealing with Indeterminate Powers:
Thus far, the only indeterminate forms we have not addressed are the indeterminate powers. For this suppose that $\lim_{x\to a} f(x)^{g(x)}$ is of an indeterminate form (where $f(x)^{g(x)}$ is defined near *a*).

For our discussion below we use the notation $\exp\left(f(x)\right)=e^{f(x)}$ (this notation is very common in differential equations)

The key to finding limits of this form is to notice that $f(x)^{g(x)}=\exp\left(\ln\left(f(x)^{g(x)}\right)\right)=\exp\left(g(x)\ln\left(f(x)\right)\right)$.

Example 7: Find $\lim_{x\to 0^+} x^x$.

We notice that this limit has the indeterminate form of 0^0. However we cannot yet apply L'Hopitals rule because we first need the indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We note that $\lim_{x\to 0^+} x^x = \lim_{x\to 0^+} \exp\left(\ln\left[x^x\right]\right) = \lim_{x\to 0^+} \exp\left(x\ln(x)\right) = \exp\left(\lim_{x\to 0^+}\left[x\ln(x)\right]\right) = \exp\left(\lim_{x\to 0^+} \frac{\ln(x)}{\left(\frac{1}{x}\right)}\right)$. This limit has the indeterminate form of $\frac{0}{0}$, so we can apply L'Hopital's rule.

$$\exp\left(\lim_{x\to 0^+}\frac{\ln(x)}{\left(\frac{1}{x}\right)}\right) \overset{H}{=} \exp\left(\lim_{x\to 0^+}\frac{\left(\frac{1}{x}\right)}{\left(\frac{-1}{x^2}\right)}\right) = \exp\left(\lim_{x\to 0^+} -x\right) = \exp(0) = e^0 = 1.$$

Therefore we can say that $\lim_{x\to 0^+} x^x = 1$.

Example 8: Find $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$.

We notice that this limit has the indeterminate form of 1^∞. However we cannot yet apply L'Hopitals rule because we first need the indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

We note that $\lim_{x\to\infty}\exp\left(\ln\left[\left(1+\frac{1}{x}\right)^x\right]\right)=\exp\left(\lim_{x\to\infty}x\cdot\ln\left(1+\frac{1}{x}\right)\right)=\exp\left(\lim_{x\to\infty}\frac{\ln\left(1+\frac{1}{x}\right)}{\left(\frac{1}{x}\right)}\right)$. This limit has the indeterminate form of $\frac{0}{0}$, so we can apply L'Hopital's rule.

$$\overset{H}{=}\exp\left(\lim_{x\to\infty}\frac{\left(\frac{1}{1+\frac{1}{x}}\right)\cdot\left(-\frac{1}{x^2}\right)}{\left(-\frac{1}{x^2}\right)}\right)=\exp\left(\lim_{x\to\infty}\left(\frac{1}{1+\frac{1}{x}}\right)\right)=\exp\left(\frac{1}{1+0}\right)=\exp(1)=e^1=e.$$

Therefore we can say that $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=e$.